



PYTHON PROGRAMMING

Y8 Python Programming

Lesson Question	How can I store data?
Topic Question	How do I program a computer?

Y8 - Programming - Python



How do I program a computer?

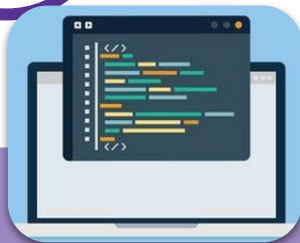


**Outwood
Travels**

Assessment

Feedback

How can I
create a simple
calculator in
Python?

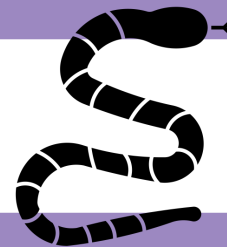


What code lets a computer
perform a calculation?

How can a computer respond
to what I tell it to do?

How can a computer
make decisions?

Where can I store
data in a program?



Theory

How do I
output data?





Recap & Recall

The code below prints "This is how you write words on the screen"

```
print("This is how you write words on to the screen")
```

Write the code that will output your name onto the screen, then today's date on the line below.

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Lesson Question

How can I store data?

Topic Question

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Keywords

Data is a piece of information, such as a number or a name.

A **variable** is how a program stores data for later use.

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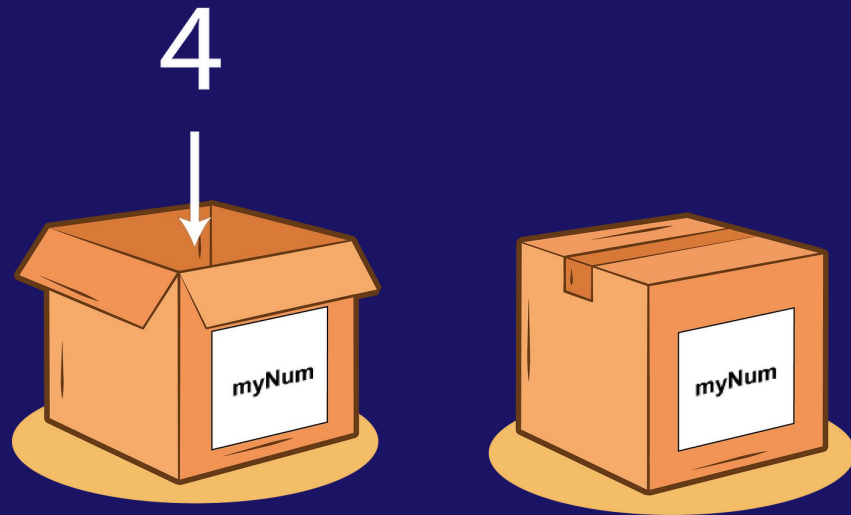


Variables

You can think of variable like a box, which has a label.

The code to the right creates a variable called myNum and puts the number 4 into it.

```
myNum = 4
```



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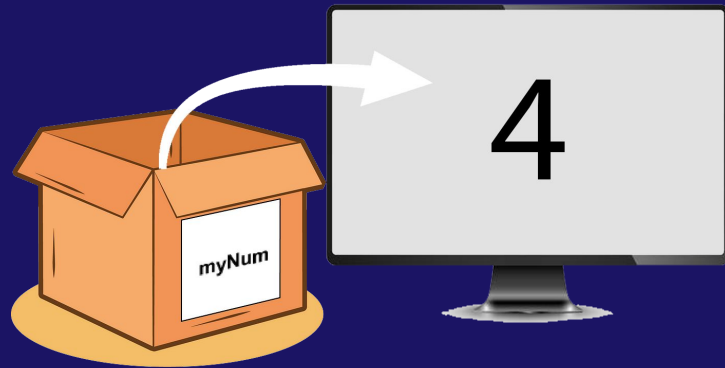


Variables

```
print(myNum)
```

You can then print what is in a variable to the screen.

This is like taking it out of the box again.



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Variables

Discuss in your row - what will the following code do?

```
number1 = 5  
number2 = 3  
print( number1 + number2 )
```

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Run Your Code

Open 2. Variables - Student Evidence - Task 1

Run the code and check to see if your prediction was correct

```
number1 = 5  
number2 = 3  
print( number1 + number2 )
```

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Investigate Your Code

- Go to task 2.
- Answer the questions about the code you have just run. You have ~4 minutes
- Make sure you are giving detail in your answer, and typing in full sentences.

Investigate

Task 2

Answer the following questions:

Question	Type your answer here. Do not give one or two word answers.
Was your prediction correct? What happened when you ran your code?	
What happens if you change the variable name from <u>number1</u> to num1? You will need to do this in two places.	
What happens if you put a space between num and 1? i.e. num 1 = 5	
Fix any errors. Then what happens if you put speech marks around the "5" and the "3"? Why does this behave differently?	
Extension: What happens if you use a variable that hasn't been created? What error do you get?	

Lesson Question**How can I store data?****Topic Question****How do I program a computer?**



Investigate Your Code

What were your answers to each question?

Investigate





Task 2

Answer the following questions:

Question	Type your answer here. Do not give one or two word answers.
Was your prediction correct? What happened when you ran your code?	
What happens if you change the variable name from <u>number1</u> to num1? You will need to do this in two places.	
What happens if you put a space between num and 1? i.e. num 1 = 5	
Fix any errors. Then what happens if you put speech marks around the "5" and the "3"? Why does this behave differently?	
Extension: What happens if you use a variable that hasn't been created? What error do you get?	

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Modify Your Code

Got to task 3.

- Change the names of the variables from number1 and number2 to num1 and num2
- Add more print statements to the bottom of the code so it shows the numbers:
 - Added together
 - Subtracted from each other (each way)
 - Multiplied by each other. (multiply by using * not x)
- For example, if my numbers were 3 and 6 the output would be:

```
9
-3
3
18
```

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Make Something New

Got to task 4.

- Create a variable called name and set it to your name.
- Create a variable called age and set it to your age.
- Print out your age multiplied by 3.

Extension: Create three more variables about yourself and print them all out in one line. Look at task 5 for how to do this.

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Predict

We can combine variables and the print statement to get neatly formatted output. What do you think this code will do?

```
name = "Riley"
mood = "Happy"

print("You are", name, "and you are feeling", mood)
```

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Run Your Code

Go to task 5 and run the code.

```
name = "Riley"  
mood = "Happy"  
  
print("You are", name, "and you are feeling", mood)
```

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Investigate Your Code

- Go to task 6.
- Answer the questions about the code you have just run. You have ~4 minutes
- Make sure you are giving detail in your answer, and typing in full sentences.

Investigate

Task 6

Answer the following questions:

Question	Type your answer here. Do not give one or two word answers.
Was your prediction correct? What happened when you ran your code?	
How many variables are in the program?	
What are the names of the variables?	
What does the comma do in the print statement?	
Why does there need to be a comma after name, but not after mood?	

Extension: What happens if you switch the comma for a plus sign? Google the difference between the two

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Investigate Your Code

What were your answers to each question?

Investigate

Task 6

Answer the following questions:

Question	Type your answer here. Do not give one or two word answers.
Was your prediction correct? What happened when you ran your code?	
How many variables are in the program?	
What are the names of the variables?	
What does the comma do in the print statement?	
Why does there need to be a comma after name, but not after mood?	
Extension: What happens if you switch the comma for a plus sign? Google the difference between the two	

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Modify Your Code

Go to task 7. Modify your code so it is showing your name and your current mood.

Create an age variable, the change the print statement so it would give the following sentence using the age variable.

Take a screenshot of this code and put it into your evidence diary.

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Make Something New

Go to task 8. Create a program that stores six different variables which describe a person - e.g. name, favouriteFood, age

Output two sentences that use these variables.

Change the content of the variables for different sentences.

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Plenary

Write into your planners that Python Homework 2 is due **next lesson**.

Then on your MWB, write a line of code that will store your name as a variable. Use it to print your name onto the screen.

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